Exercises 5.3 7, 11, 13, 17, 21, 25, 29, 31, 37, 41, 43, 47, 61, 67, 75.

7–8 Sketch the area represented by $g(x)$. Then find $g'(x)$ in two ways: (a) by using Part 1 of the Fundamental Theorem and (b) by evaluating the integral using Part 2 and then differentiating.

7. $g(x) = \int_1^x t^2\,dt$

a) $\frac{d}{dx}\int_1^x t^2\,dt = t^2$ x^2

b) $\int_1^x t^2\,dt = \left[\frac{t^3}{3}\right]_1^x$

$= \frac{x^3}{3} - \frac{1}{3}$

$\frac{d}{dx}\left(\frac{x^3}{3} - \frac{1}{3}\right) = x^2$

8. $g(x) = \int_0^x (2 + \sin t)\,dt$

a) $\frac{d}{dx}\left[\int_0^x (2+\sin t)\,dt\right] = 2+\sin x$ $2+\sin t$

b) $\int_0^x (2+\sin t)\,dt = 2x - \cos t$

$\frac{d}{dx}(2x - \cos t) = 2+\sin t$

$\int_0^x (2+\sin t)\,dt = [2x - \cos t]_0^x$

$= 2x - \cos x - 1$

$\frac{d}{dx}(2x - \cos x - 1) = 2 - (-\sin x) - 0$

$= 2 + \sin x$

9–20 Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of the function.

11. $g(w) = \int_0^w \sin(1 + t^3)\, dt$

$\frac{d}{dw}\int_0^w \sin(1+t^3)\,dt = \sin(1+t^3)$

13. $F(x) = \int_x^0 \sqrt{1 + \sec t}\, dt$

$= -\frac{d}{dx}\int_0^x \sqrt{1+\sec t}\, dt$

FTC I $= -\sqrt{1+\sec t}$

17. $y = \int_1^{3x+2} \frac{t}{1 + t^3}\, dt$

Let $u = 3x+2$; $\frac{du}{dx} = 3$ Also $\frac{dy}{dx} = \frac{du}{dx}\cdot\frac{d}{du}$

$= \frac{du}{dx}\cdot\frac{d}{du}\int_1^u \frac{t}{1+t^3}\, dt$

$= 3\cdot\frac{u}{1+u^3}$

$= 3\cdot\frac{3x+2}{1+(3x+2)^3}$

$y' = \frac{d}{dx}\int_1^{3x+2} \frac{t}{1+t^3}\, dt$

$= \frac{du}{dx}\cdot\frac{d}{du}\cdot\int_1^u \frac{t}{1+t^3}\, dt$

$= 3\cdot\frac{u}{1+u^3}$

21–24 Use Part 2 of the Fundamental Theorem of Calculus to evaluate the integral and interpret the result as an area or a difference of areas. Illustrate with a sketch.

21. $\int_{-1}^{2} x^3\,dx$

22. $\int_{0}^{4} (x^2 - 4x)\,dx$

29. $\int_{1}^{9} \sqrt{x}\,dx$

31. $\int_{0}^{4} (t^2 + t^{3/2})\,dt$

37. $\int_{1}^{4} \dfrac{2 + x^2}{\sqrt{x}}\,dx$

41. $\int_{0}^{\pi/3} \sec\theta \tan\theta\,d\theta$

43. $\int_0^1 (1+r)^3\,dr$

$= \frac{1}{4}\left[(1+r)^4\right]_0^1$

$= \frac{1}{4}\left[2^4 - 1\right]$

$= \frac{1}{4} \times 15 = 3.75$

47. $\int_0^1 (x^e + e^x)\,dx$? 真笑e的个指数

$= \left[\frac{x^{e+1}}{e+1} + e^x\right]_0^1$

$= \frac{1}{e+1} + e - (0+1)$

$= \frac{1}{e+1} + e - 1$

67–71 Find the derivative of the function.

67. $g(x) = \int_{2x}^{3x} \frac{u^2-1}{u^2+1}\,du$

$$\left[\textit{Hint: } \int_{2x}^{3x} f(u)\,du = \int_{2x}^{0} f(u)\,du + \int_{0}^{3x} f(u)\,du\right]$$

$g(x) = \int_{2x}^{0} f(u)\,du + \int_{0}^{3x} f(u)\,du$

$= -\int_{0}^{2x} f(u)\,du + \int_{0}^{3x} f(u)\,du$ → $u_2 = 3x$, $\frac{du}{dx} = 3$, Also, $\frac{d}{dx} = \frac{du}{dx}\cdot\frac{d}{du}$

$u_1 = 2x$, $\frac{du}{dx} = 2$, Also $\frac{d}{dx} = \frac{du}{dx}\cdot\frac{d}{du}$

$\frac{d}{dx}g(x) = -\frac{4x^2-1}{4x^2+1} + \frac{9x^2-1}{9x^2+1} \times$

$= \frac{du_1}{dx}\cdot\frac{d}{du_1}\left(-\int_0^{u_1} \frac{u^2-1}{u^2+1}\,du\right) + \frac{du_2}{dx}\cdot\frac{d}{du_2}\left(\int_0^{u_2} \frac{u^2-1}{u^2+1}\,du\right)$

$= 2\cdot\frac{4x^2-1}{4x^2+1} + 3\cdot\frac{9x^2-1}{9x^2+1}$

75. Let $F(x) = \int_2^x e^{t^2}\,dt$. Find an equation of the tangent line to the curve $y = F(x)$ at the point with x-coordinate 2. $x = 2$

$F(x) = \left[\frac{e^{t^2}}{2t}\right]_2^x =$

$\frac{d}{dx} F(x) = F'(x) = e^{t^2}$

$F'(2) = e^4$ $\Rightarrow$ slope when $x = 2$

$F(2) = \int_2^2 e^{t^2}\,dt = 0$ since the limits are equal

2. the tangent line is $y - 0 = e^4(x - 2)$